

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY,
BHOPAL

NETWORK LAB
EXPERIMENT NO.06

Aim: To study the ABCD parameters of a two port network.

Apparatus Required:

1. Trainer kit
2. Connecting cords

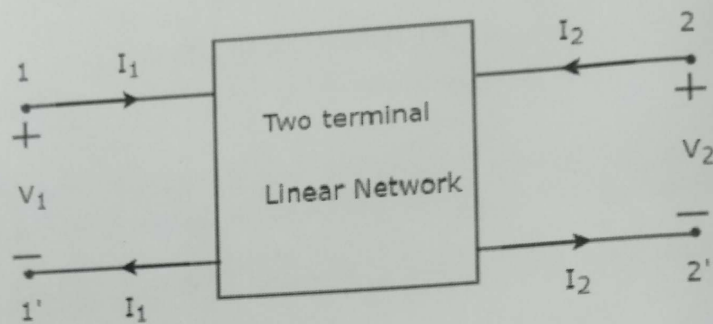
Theory:

A pair of terminals at which a signal (voltage or current) may enter or leave is called a port. A network having only one such pair of terminals is called as one port network. A two-port network (or four-terminal network, or quadri-pole) is an electrical circuit or device with two pairs of terminals.

Examples include transistors, filters and matching networks. The analysis of two-port networks was pioneered in the 1920s by Franz Breisig, a German mathematician.

A two-port network basically consists in isolating either a complete circuit or part of it and finding its characteristic parameters. Once this is done, the isolated part of the circuit becomes a "black box" with a set of distinctive properties, enabling us to abstract away its specific physical buildup, thus simplifying analysis. Any circuit can be transformed into

a two-port network provided that it does not contain an independent source.



A two-port network is represented by four external variables: voltage and current at the input port, and voltage and current at the output port, so that the two-port network can be treated as a black box modelled by the relationships between the four variables, and there exist six different ways to describe the relationships between these variables, depending on which two of the four variables are given, while the other two can always be derived.

Note: All voltages and currents below are complex variables and represented by phasors containing both magnitude and phase angle. However, for convenience the phasor notation V and I are replaced by V and I respectively.

The parameters used in order to describe a two-port network are the following: Z , Y , A , h , g . They are usually expressed in matrix notation and they establish relations between the following parameters:

Input voltage	V_1
Output voltage	V_2
Input current	I_1
Output current	I_2

ABCD-model: In the A-model or transmission model, we assume V_1 and I_1 are known, and find V_2 and I_2 by:

$$\begin{pmatrix} V_2 \\ I_2 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} V_1 \\ I_1 \end{pmatrix}$$

Where,

$$A = \frac{V_2}{I_2} \text{ when } I_2 \text{ is zero}$$

$$B = \frac{V_2}{I_1} \text{ when } V_1 \text{ is zero}$$

$$C = -\frac{I_2}{V_1} \text{ when } I_2 \text{ is zero}$$

$$D = -\frac{I_2}{I_1} \text{ when } V_1 \text{ is zero}$$

Here A and D are dimensionless coefficients, B is impedance and C is admittance. A negative sign is added to the output current I_2 in the model, so that the direction of the current is outward, for easy analysis of a cascade of multiple network models.

Procedure :

- Connect the circuit as shown in connection diagram shown below.
- While making the connections make sure that connections are correct.
- Connect all meters with correct polarities.
- Switch on the trainer kit
- Set the DC power supply to 5V provided on the panel.

PART A:

1. First open the input terminal and connect the power supply 5V to the output terminal. Measure the output voltage (V_2), input voltage (V_1) and output current (I_2).
2. Note the reading in the tabular column as shown below.
3. Repeat the procedure for another set of reading by varying the dc power supply provided on the panel.

PART B:

1. Secondly short the input terminals and connect the supply 5V to the output terminal. Measure the input current (I_1), output current (I_2) and output voltage (V_2) with the meter provided.
 2. Record the reading obtained as per in the table given below.
 3. Repeat the procedure for another set of readings.
- Calculate the ABCD parameters using the formula.
 - Switch off the trainer after taking the readings.

Observation Table :

S.NO.	When input is open circuited (i.e. $I_2 = 0$)			When input is short circuited (i.e. $V_1 = 0$)		
	V_1	V_2	I_1	V_2	I_2	I_1
1.						
2.						
3.						
4.						
5.						

Calculations :

$$A = \frac{V_2}{I_2} \text{ when } I_2 \text{ is zero}$$

$$B = \frac{V_2}{I_1} \text{ when } V_1 \text{ is zero}$$

$$C = -\frac{I_2}{V_1} \text{ when } I_2 \text{ is zero}$$

$$D = -\frac{I_2}{I_1} \text{ when } V_1 \text{ is zero}$$

Tabulation :

S.No.	Parameter	Theoretical Value	Practical Value
1.	A		
2.	B		
3.	C		
4.	D		

Result: If Practical and theoretical values meet near approximation. Then required result is verified.

Questions

1. What is the two port network?
2. If two-port networks are connected in cascaded form, then which two port network parameters can be multiplied?
3. Which matrices of two port network parameters are inverse to each other?
4. If two port networks are connected in series, then which two port network parameters can be added?
5. If two port networks are connected in parallel, then which two port network parameters can be added?

ANS.

1. A two-port network consists of two ports. So, there are four variables V_1 , V_2 , I_1 and I_2 . At any time, by choosing two of them as independent variables, we will get a total of six sets of parameters regarding the two port networks. The two-port network parameters are Z , Y , T , T' , h , and g .
2. Cascade means the output of one stage we will give as the input for the next stage. If multiple blocks are connected in cascaded form, then we will get the equivalent block transfer function by multiplying the transfer function of each block. Similarly, if the two-port networks are connected in cascaded form, then we can multiply the respective T parameters or Transmission parameters of those networks.
3. We are having a total of six sets of two-port network parameters Z , Y , T , T' , h and g . Among which the pair of matrices of two port network parameters, which are inverse to each other are (Z, Y) , (T, T') , and (h, g) . That is the matrices $[Z]$ and $[Y]$ are inverse to each other. Similarly, the matrices $[T]$ & $[T']$, $[h]$ & $[g]$ are inverse to each other.
4. If multiple resistors are connected in series, then we will get the equivalent resistance value by adding all those resistances. Similarly, if the two port networks are connected in series, then we can add the respective Z

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parameters or impedance parameters of those networks.

5. If multiple resistors are connected in parallel, then we will get the equivalent conductance value by adding all those conductance. Conductance is nothing but reciprocal resistance. Similarly, if the two-port networks are connected in parallel, then we can add the respective Y parameters or admittance parameters of those networks